BIWEEKLY INTELLIGENCE REPORT / ISSUE 02

El Niño 2026–2027: strong signal, unequal local risks

Assessment date: **23 October 2026** | Main outlook: **October 2026–March 2027** | Compared with brief of 9 October 2026

Executive brief — in plain English

El Niño is **already established and exceptionally strong**. The equatorial Pacific is unusually warm, atmospheric winds and rainfall have shifted, and all six requested forecasting institutions expect the event to remain influential into early 2027. The strongest shared forecast is for a peak near **November–December 2026**, followed by a gradual weakening—not an abrupt end. [1–6]

The key risk is **not the same everywhere**. Northern Brazil and parts of the Amazon are more exposed to a combination of rainfall deficits, heat, vegetation dryness and navigation constraints. Southern Brazil faces a contrasting wet-season/flood signal. Peru's coastal areas face increasingly wet conditions while parts of its Amazon are forecast near or below normal rainfall. These are **regional probability shifts, not guarantees** of floods, fires or colony losses. [7–11]

> **Bottom line for the Fund:** a credible, time-limited case exists for financing localized preparedness and field monitoring *before* seasonal risks intensify. Donors should fund readiness, not a falsely certain disaster prediction. [8, 9, 12]

What changed since 9 October?

- **New IRI analysis (19 Oct):** traditional Niño-3.4 weekly reading **+3.3°C** (centered 14 Oct); 27-model plume places its approximate three-month peak near **+3.5°C**. This is not directly interchangeable with NOAA's +2.1°C figure from 8 Oct. [1, 2]
- **New WMO outlook (21 Oct):** multi-model **+3.8°C forecast** for November 2026–January 2027; replaces the former October–December projection as the newest seasonal window, *not* a new observed temperature. [3, 4]
- **New Copernicus (10 Oct) and Australia (13 Oct):** both favor a late-2026 peak, but C3S flags disagreement on peak amplitude and a fairly steep decline afterwards. [5, 6]
- **New Brazil and Peru risk evidence:** ANA's 19 Oct release reports drought across **75% of Brazilian territory in September versus 67% in August**; Peru ENFEN issued revised coastal and river projections on 16 Oct. Drought percentage is *not* an estimate of El Niño-attributable drought. [9–11]
- **No new monthly NOAA diagnostic since 8 Oct;** Brazil's latest interagency El Niño Panel remains No. 4 (1 Oct). The quarterly WMO El Niño/La Niña update has not been replaced since September; its 21 Oct global seasonal forecast *is* newer. [1, 3, 7, 14]

1. Evidence: what is observed and what is forecast?

Indicator / issuer	Latest observation or forecast	Interpretation
NOAA CPC | 8 Oct	Observed weekly Niño-3.4 **+2.1°C**. El Niño Advisory. **83%** chance October–December reaches RONI ≥ +2.5°C. [1]	A high probability of *exceptional relative* strength for OND, **not** an 83% prediction of a disaster.
IRI / Columbia | 19 Oct	Traditional Niño-3.4 weekly **+3.3°C** centered 14 Oct; September monthly **+2.84°C**. [2]	Different date/data/metric from NOAA. Do not present the difference as a one-week jump.
Australia BoM | 13 Oct	Relative Niño-3.4 **+2.81°C**, week ending 11 Oct; highest weekly value in its GAMSSA series since 2008. [6]	The Bureau explicitly notes monthly values have **not** broken the longer-term record.
WMO | 21 Oct	Model-ensemble mean **+3.8°C** Niño-3.4 for Nov–Jan 2026–27. [3]	**Forecast** anomaly, not observed. Published WMO chart uses a distinct 1993–2009 comparison baseline.
IRI model plume | 19 Oct	All **27 models** project very strong traditional Niño-3.4 for OND through DJF; ensemble mean near **+3.5°C**. [2]	Strong agreement on phase; spread remains around magnitude, peak timing and spring evolution.

ENSO measures an ocean–atmosphere climate pattern. **Niño-3.4** is the warming of a defined central/eastern equatorial-Pacific zone. A **relative** index (RONI or a relative Niño-3.4 formulation) subtracts wider tropical warming to better isolate the ENSO signal. Index baselines, datasets and averaging windows differ across agencies. **+2.1, +2.81, +3.3 and +3.8°C are not a valid single time series.** [1–6]

2. Probability and disagreement, explained

Persistence: IRI's October model-based probabilities round to **100% for El Niño from OND 2026 through MAM 2027**, falling to 92% in April–June and 60% in May–July. "100%" is a rounded output of a model procedure, not a literal guarantee. NOAA's official October discussion separately states that strong-to-very-strong conditions remain above an 83% chance through January–March. [1, 2]

Peak strength: Copernicus' 10 October multi-system analysis confirms a likely peak around the turn of the year, with systems **disagreeing on maximum magnitude**. IRI's 27 models agree on very strong conditions but differ on peak heights; WMO's newer three-month mean is +3.8°C. The independent agreement is more persuasive than any individual peak estimate. [2, 3, 5]

Spring and beyond: C3S favors a relatively rapid decline after the peak, but continued El Niño conditions through its forecast horizon. IRI still favors El Niño in spring and sees ENSO-neutral becoming most likely by June–August 2027. These are compatible possibilities; neither supports claiming that harmful local effects end the moment Pacific indices start falling. [2, 5]

3. Where impacts may emerge — a regional risk register

Use three categories: **OBSERVED** (already measured), **FORECAST** (seasonal odds), and **CONDITIONAL RISK** (possible consequences if local exposure and weather align).

Geography & window	Evidence and forecast	Responsible planning implication
Northern Amazon, Brazil Oct–Dec 2026	OBSERVED: low levels in Branco/Negro systems; drought worsened in Roraima, Acre, Amazonas in August/September. FORECAST: below-normal rainfall and above-normal heat in portions of Roraima, Amapá, Amazonas and Pará. [7–9]	Check locality-level river access, safe water, burn exposure and hive-site shade. A continental Amazon emergency has **not** been established.
Amazon interior / Pará Oct–Dec 2026	FORECAST: many locations below seasonal rainfall average; parts of south/central region can still have isolated heavy storms and normal seasonal wet-season onset. [8]	Do not claim Belém, Caxiuanã, Manaus and Roraima will experience identical conditions.
South Brazil Oct 2026–Jan 2027	OBSERVED: September flooding and high river levels in Caí/Taquari and some Santa Catarina catchments. FORECAST: above-normal seasonal rainfall; flood risk tending lower toward December. [7, 8]	Prepare flood/watershed response separately from Amazon drought messaging; use current municipal alerts.
Peru north/central coast Late 2026–Mar 2027	ENFEN (16 Oct) forecasts increasingly above-normal rains; named rivers Tumbes, Piura, Chira and others may have higher flows Nov onward. [10, 11]	Communicate flood preparedness as a *prospect*, not a dated flood prediction.
Peruvian Amazon Late 2026–early 2027	FORECAST: near- to below-normal rain, with below-normal flows expected in Ucayali and Huallaga and parts of central/southern Amazon. [11]	Assess water access, transport and forest conditions; local satellite/river data must guide operations.
Central America / Caribbean Oct 2026–early 2027	WMO identifies seasonal dry signals in southern Central America and parts of the Caribbean; 22 Oct humanitarian update emphasizes risk rather than verified outcomes. [4, 12]	Where active, pair ENSO information with national meteorological and disaster-management notices.

4. Critical Brazil observations — what they do and do not prove

Brazil's ANA Monitor de Secas, released **19 October**, assessed September drought over **75% of Brazil**, up from 67% in August. Severity worsened in **14 states**, including Acre, Amazonas, Amapá, Pará and Roraima; all three southern states remained drought-free. Acre's **severe drought extent rose from 37% to 53%** of its area. These are monitoring classifications of accumulated conditions, not proof that El Niño caused each affected area. [9]

The INPE-led September Panel is careful: current Amazon hydrology is **heterogeneous**; it does **not** identify a basin-wide extreme low-water emergency like 2023–24. It calls for closer attention in Branco/Negro basins. The same panel's seasonal CENSIPAM outlook forecasts lower precipitation in much of northern Amazonia and notes possible delayed rains in far-northern areas. [8]

5. Stingless bees and forest-dependent communities

Observed scientific foundation: stingless-bee species differ in their thermal tolerance, forage patterns and ability to survive dry periods. An open university-hosted field study of *Melipona subnitida* in Brazil's Caatinga found that colonies use diverse native flowering plants across seasons, including dry periods. It supports protecting local floral resources—but **does not measure 2026 El Niño effects** in Amazonian *Melipona* or *Tetragonisca*. [13]

Reasonable conditional pathways: high heat can affect foraging schedules and nesting conditions; altered rainfall may shift flowering and the availability of nectar/pollen; smoke and low river levels may interrupt field access and livelihoods. Because rainfall can be localized and species differ, none of these constitutes a demonstrated El Niño-caused bee mortality forecast. [8, 13]

Recommended low-regret field evidence, October–March

- **Community observations:** simple weekly logs for colony activity, flowering plants, available forage and local weather, with informed consent.
- **Habitat measures:** locally approved shade, firebreak and native-flowering-plant protection; avoid habitat manipulation without ecological review.
- **Access and water:** practical monitoring of community water reliability, smoke alerts and navigability, using Brazil and Peru's official bulletins.
- **Documentation:** record baseline photos/field notes before assistance and follow up afterwards; report both success and non-results.

6. Fundraising and editorial timing

Now–mid-November: prioritize preparedness budgets, partner needs assessments and a specific, independently checkable use-of-funds plan. WMO's 22 October humanitarian communication explicitly supports advance action while forecasts still provide lead time. Use this as policy context, not as an endorsement of the Fund. [12]

Late November–January: report observed local conditions alongside model forecasts; release photographs, river measurements and community interviews only with adequate consent and date/location labels. Establish an emergency reserve only for verified triggers. **February–March:** assess impacts and publish outcome accounting, even if the worst impacts do not occur.

Five donor-facing talking points — cleared for evidence-grounded use

1. "El Niño is already strong. The world's leading forecast centers expect its influence to continue into early 2027; the locations and severity of impacts will vary."
2. "In September, Brazil's drought monitor classified 75% of the country as affected by drought. That is an observed monitoring result—not proof that El Niño caused every drought."
3. "Official Brazilian outlooks identify elevated dry-season concern in parts of northern Amazonia, while southern Brazil faces a different, wetter risk pattern."
4. "Stingless bees depend on healthy, diverse flowering habitats. We are preparing to document local conditions and support community-defined adaptation, not claiming predicted mass bee deaths."
5. "Forecasts buy time. Support now can finance practical monitoring, habitat care and community preparedness before the peak-risk months."

Social-post controls

Always label “observed,” “forecast,” and “possible”; date every statistic; name the specific region; link to the original public release. Never say “guaranteed floods,” “the worst ever” as an observed outcome, or “El Niño killed bees” without field evidence. Use WMO/NOAA maps only with source labeling and applicable credit. WMO does not endorse the phrase “Super El Niño.” [1–5, 14]

7. Primary-source access and freshness ledger

Each link below was opened and its readable content inspected on **23 October 2026** through the available web access. No login or paywall was encountered for the listed landing pages or the two linked primary PDFs. Actual behavior can vary by device, geography, future site changes and security settings. All source names are active hyperlinks in this PDF.

[01] NOAA CPC ENSO Diagnostic Discussion — 8 Oct 2026

[02] IRI / Columbia ENSO Quick Look & model plume — 19 Oct 2026

[03] WMO Global Seasonal Climate Update: NDJ 2026-27 — 21 Oct 2026

[04] WMO GSCU briefing: OND 2026 — 8 Oct 2026

[05] Copernicus C3S multi-system seasonal highlights — 10 Oct 2026

[06] Australian Bureau of Meteorology: Southern Hemisphere Monitoring — 13 Oct 2026

[07] INPE / interagency Brazil El Niño Panel No. 4 — 1 Oct 2026

[08] INPE/INMET/CENSIPAM panel: technical bulletin (PDF) — Sep 2026; posted 1 Oct

[09] Brazil ANA: Monitor de Secas, September assessment — 19 Oct 2026

[10] Peru ENFEN: Official Bulletin No. 18 (landing) — 16 Oct 2026

[11] Peru ENFEN: Official Bulletin No. 18 (PDF) — 16 Oct 2026

[12] WMO: humanitarian weather services and anticipatory action — 22 Oct 2026

[13] UNESP repository: Melipona subnitida drought/forage study — Peer-reviewed; repository

[14] WMO El Niño / La Niña Update hub — Quarterly, latest Sep 2026

Interpretation and safeguards

Verification scope: the published pages/PDFs opened, and the scientific claims used here were checked against their own text. Model forecast probabilities do not translate directly into percentages for drought, fire, flood, livelihood loss or pollinator mortality. Agency indices differ in reference periods, datasets, weekly/monthly/three-month averages and traditional versus relative calculations. Forecast weather risk is not causal attribution for observed losses.

Source recency: NOAA CPC’s latest monthly diagnostic is 8 Oct (next scheduled 12 Nov). IRI has a new 19 Oct plume. WMO released an NDJ seasonal outlook 21 Oct and a humanitarian update 22 Oct; the quarterly ENSO-specific update remains September. Copernicus seasonal highlight is 10 Oct, Australia’s monitoring is 13 Oct, and Brazil’s INPE-led El Niño Panel No. 4 dates to 1 Oct. The latest ANA drought assessment is 19 Oct. Peru ENFEN Bulletin 18 is 16 Oct (next stated bulletin 30 Oct). [1–12, 14]

Editorial independence: inclusion of agencies or scientific institutions as information sources does not imply they endorse, partner with or fund the EL NIÑO RESILIENCE FUND. Any financial solicitation should separately document the Fund’s legal status, specific activities, budget and reporting commitments.

Monitoring priority for the next edition: NOAA’s 12 November diagnostic; the next IRI and C3S model updates; Peru ENFEN’s 30 October release; Brazil Panel No. 5 and additional ANA river and drought observations. Track *changes in location-specific exposure* before changing appeals.